IPR 2 Decision Brief

Purpose: Confirm the Adelman mathematics, course-requirement coverage, and evidence plan for the final presentation.

Decisions requested

- **Slide 8:** Proof and LP depth?
- **Slide 10:** IPR 3 computation scope?
- **Slide 11:** Published replication target?
- **Slide 13:** Conceptual application scope?

Agenda today

- Progress and requirement crosswalk
- Exact, fluid, and affine models
- Bound proof and column generation
- Verification and experiment design
- Final-brief scope and risks

Successful end state

- Theory coverage agreed
- Numerical gates defined
- IPR 3 comparisons approved
- Final narrative ordered
- Extension limits recorded

Source: Structure follows the IPR 1 decision brief, p. 1. Status and requests are project-specific.

Final-presentation crosswalk and current status

Section	Deliverable	Status	Next action
1. Intro (I14)	Pricing motivation and approved RQs	Draft	Verify history and prevalence
2. RM (I01–I03)	Discounted LP, exact MDP, fluid LP	Draft	Audit signs and example
3. Method (I04–I06, I09)	Affine ALP, duality, decision rule	Draft	Complete formal audit
4. Proof (I07, I10)	Bound hierarchy and theorem	Draft	File independent proof audit
5. Scale (I11)	Pricing and column generation	Draft	Verify complexity and separation
6. Design (I08, I12)	Benchmarks, metrics, policies	Draft	Approve factors and baselines
6. Results (I08, I12)	Code, tables, chart, sensitivity	Not started	Pass preflight and tests
7. Critique (I13)	Assumptions and alternatives	Draft	Link claims to evidence
8. Literature (I14)	Pricing lineage and FTC context	Not started	Verify works and primary source
8. Extension (I14)	Synthetic pricing application	Blocked	Await course and human gate
9. Conclusion	Three supported takeaways	Draft	Update after results

Key: Red Blocked Yellow Draft Amber Not started Green Complete

Source: User's final outline and project phase plan. Status describes presentation progress; formal requirements I01–I14 remain unverified in the project register.

Roadmap from IPR 1 to the final presentation

IPR 1: scope	IPR 2: theory	IPR 3: evidence	Final: synthesis
• Six approved questions • LP themes • Verification plan • Application guardrails	• Exact and discounted LPs • Fluid and affine models • Bound proof • Pricing formulation • Experiment approval	• Independent oracle • Certified bounds • Static/dynamic policies • Scaling and truncation • Reviewed results	• Motivation • Adelman result • Measured comparisons • Critique • Gated application • Limits

Current gate: Theory drafted. Formal audit, Windows solver preflight, code, and numerical results pending.

Final brief rule: every numerical chart must trace to a reviewed result. The optional application cannot displace instructor-required theory or computation.

Source: Roadmap structure follows IPR 1 brief, p. 6; milestones follow the project phase plan.

Research scope and current progress

Question groups	
Theory (RQ1–3)	What does the affine value approximation retain? Why is its bound stronger? How do dual prices and column generation work?
Computation (RQ4–5)	How do size and truncation affect cost and quality? When do dynamic prices improve expected reward?
Application (RQ6)	Does the resource-allocation model transfer to surveillance missions?

Formulations and proofs	Source analysis drafted; formal audit artifact pending.
Implementations and tests	Exact DP, LPs, policies, and assertions pending.
Experiments and replication	Design proposed; no project run or published-table reproduction.

Source: Six approved research questions and the project phase plan.

Finite-horizon network revenue management

State	Period $t \in \{1, \dots, T\}$ and remaining resource vector $r \in \mathcal{R} = \prod_{i=1}^{m} \{0, \dots, c_i\}$.
Arrival	Class j arrives with probability p_{tj}; no arrival has probability $1 - \sum_j p_{tj}$. At most one request arrives per period.
Action	$u_j \in \{0, 1\}$ says whether class j would be accepted; $a_{ij} u_j \le r_i$ for every resource i and class j.
Reward and next state	An accepted class j earns f_j and moves to $r - A_j$; otherwise capacity stays at r.

One-period MDP sketch

$$\text{state } (t, r) \;\downarrow\; \text{class } j \text{ arrives with } p_{tj} \;\begin{cases} \text{accept: } (f_j,\, t+1,\, r - A_j), \\ \text{reject: } (0,\, t+1,\, r). \end{cases}$$

Adelman's acceptance rule

$$u^*_{tj}(r) = 1 \quad \Longleftrightarrow \quad r \ge A_j, \quad f_j - \Big[v_{t+1}(r) - v_{t+1}(r - A_j) \Big] \ge 0.$$

Revenue today must cover the next-period value of capacity used.

Variables: $t = 1, \dots, T$ period; $i = 1, \dots, m$ resource; $j = 1, \dots, n$ class; c_i initial and r_i remaining capacity; a_{ij} units used; $A_j = (a_{ij})_i$; p_{tj} arrival probability; u_j accept indicator; f_j fare; $v_t(r)$ optimal future reward.

Source: Adelman (2007), pp. 648–649, §2.1 and eqs. (1)–(2).

Exact MDP and its linear program

Bellman recursion

$v_{T+1}(r) = 0$ and

$$v_t(r) = v_{t+1}(r) + \sum_{j: r \geq A_j} p_{tj} \left[f_j - \Delta_{t+1,j}(r) \right]_+,$$

$$\Delta_{t+1,j}(r) = v_{t+1}(r) - v_{t+1}(r - A_j).$$

For a feasible arrival, accept if

$$f_j \geq v_{t+1}(r) - v_{t+1}(r - A_j).$$

Bellman LP (D0)

Minimize $v_1(c)$ over unrestricted $v_t(r)$ subject to

$$v_t(r) \geq q_t(r, u) + \mathbb{E}[v_{t+1}(R_{t+1}) \mid r, u]$$

for every $t, r, u \in \mathcal{U}(r)$.

Its exact dual conserves probability at every time-state pair. Complementary slackness identifies optimal actions on occupied states.

Variables: t period; T horizon; r remaining capacity; c initial capacity; A_j class-j resource use; p_{tj} arrival probability; f_j fare; $v_t(r)$ optimal value; Δ opportunity cost; $[x]_+ = \max(0, x)$; u feasible action in $\mathcal{U}(r)$; q_t expected reward; R_{t+1} next state; $\mathbb{E}$ expectation.

Source: Adelman (2007), p. 649, eqs. (1)–(3); occupancy dual independently derived in backup slide A2.

Fluid LP and affine approximate LP

Fluid LP: expected acceptances Y_j

$$\max_{Y \geq 0} \sum_j f_j Y_j \quad \text{s.t. } AY \leq c, \quad Y_j \leq \sum_t p_{tj}.$$

Resource dual prices $\lambda_i \geq 0$ give the static test

$$f_j \geq A_j^\top \lambda.$$

This solve aggregates demand over time.

Affine value approximation

$$\hat{v}_t(r) = \theta_t + b_t^\top r.$$

Substitution in (D0) gives (D1):

$$\theta_t - \theta_{t+1} + (b_t - b_{t+1})^\top r \geq \sum_j p_{tj}(f_j - b_{t+1}^\top A_j)u_j.$$

The policy accepts a feasible class j if $f_j \geq b_{t+1}^\top A_j$.

Within one affine solve, bid prices vary by period. Re-solving can update the trajectory after capacity changes.

Variables: Y_j expected acceptances; $A = (a_{ij})$ resource-use matrix; c initial capacity; f_j fare; p_{tj} arrival probability; λ_i fluid resource dual; $\hat{v}_t$ affine value; θ_t intercept; b_t resource-price vector; r remaining capacity; u_j accept indicator; t period.

Source: Adelman (2007), p. 650, eqs. (4)–(6), and p. 652, eqs. (14)–(17).

Two upper-bound arguments

$$z^{LP} \;\geq\; z^{P1} \;\geq\; v_1(c)$$

1. **Affine LP versus exact MDP.** Every feasible (D1) function satisfies all Bellman inequalities. Backward induction makes it an upper bound on $v_t(r)$. Restricting the value-function family cannot lower the minimum below $v_1(c)$.
2. **Fluid LP versus affine LP.** Given feasible (P1) weights X, set $Y_j = \sum_{t,r,u} p_{tj} u_j X_{t,r,u}$. P1's mass and resource-moment equations imply $Y_j \leq \sum_t p_{tj}$ and $AY \leq c$. The two objectives agree.

Structural result: under $c_i > 0$ and $p_{tj} > 0$, a monotone optimal trajectory $b_{t,i} \geq b_{t+1,i}$ exists. An arbitrary optimal solver output need not be monotone.

★ **Decision:** Is this theorem and proof depth sufficient for the final brief?

Variables: z^{LP} fluid upper bound; z^{P1} affine-LP upper bound; $v_1(c)$ exact initial MDP value; $X_{t,r,u}$ P1 state-action weight; Y_j expected class-j acceptance; p_{tj} arrival probability; u_j accept indicator; A resource-use matrix; $b_{t,i}$ resource-i slope; c_i initial capacity.

Source: Adelman (2007), pp. 652–655, Theorems 1–2 and eqs. (22)–(29).

Published example and planned verification

Example 1: one resource, one seat, one class, $f = a = 1$, $T = 2$, and $p_1 = p_2 = 0.5$.

$$v_2(1) = 0.5, \qquad v_1(1) = 0.5 + 0.5(0.5) = 0.75.$$

Formulation	Value	Evidence status
Exact MDP	0.75	Hand calculation; published example
Affine LP (P1)	0.75	Published example; reproduction planned
Fluid LP	1.00	Hand calculation; published example

Verification gate: independently compare exact DP, exact LP primal/dual, fluid LP, and full affine LP on this case and a two-resource network. Test zero demand, zero capacity, terminal decisions, and infeasible arrivals.

Variables: T horizon; f fare; a resource use; p_t arrival probability; $v_t(r)$ exact value at period t with remaining capacity r; P1 affine approximation; fluid LP deterministic relaxation.

Source: Adelman (2007), p. 653, Example 1. Project-generated solver results are pending.

Column generation and its certificate

Primary: start (P1) with feasible no-acceptance columns. Solve the restricted primary and obtain θ_t, b_t.

Pricing: for each period maximize reduced profit over feasible (r, u):

$$\rho_t(r,u) = \sum_j p_{tj}(f_j - b_{t+1}^\top A_j)u_j - (b_t - b_{t+1})^\top r - \theta_t + \theta_{t+1}.$$

Add a column when its reduced profit is positive, then re-solve. Adelman formulates the period problem as an integer program.

Stopping certificate (Proposition 3):

$$Z_\Lambda \le z^{P1} \le Z_\Lambda + \sum_{t=1}^{T} \rho_t^*, \quad \rho_t^* = \max_{r,u} \rho_t(r,u).$$

Z_Λ alone does not certify an upper bound on the exact MDP.

★ **Decision:** How much of the pricing and column-generation algorithm must IPR 3 implement?

Variables: t, T period/horizon; (r, u) state/action; p_{tj}, f_j, A_j arrival probability, fare, resource use; b_t, θ_t slopes/intercept; ρ_t, ρ_t^* reduced profit/period maximum; Z_Λ, z^{P1} restricted/full primary objectives.

Source: Adelman (2007), pp. 650–651, Proposition 3; pp. 655–657, §4.4 and eqs. (33)–(36).

Proposed experiment design for IPR 3

Factors	Resources, horizon, load, fare spread, arrival timing, and single- versus multi-resource products. Start with exactly enumerable cases.
Comparators	Exact DP where tractable; fluid and certified affine bounds; fixed static and dynamic policies; separately labeled re-solving variants; proposed first-come feasible baseline.
Outcomes	Bound gaps, exact or estimated policy reward, paired dynamic-minus-static difference, runtime, memory, iterations, and generated columns.
Inference	Common arrival paths and predeclared seeds; report 95% intervals for each policy mean and for paired differences. Use absolute gaps when exact value is zero.

Truncation study: record which state, column, or horizon restriction is used and whether a valid pricing certificate survives.

★ **Decision:** Which published result and comparison protocol should IPR 3 target?

Source: Proposed design; no project experiment has run. Adelman (2007), pp. 657–660, supplies comparison context.

Critical appraisal for the final presentation

Model assumption	Question to test or disclose
Known exogenous arrivals	How sensitive are policies to misspecified probabilities?
Nonreplenishing capacity	Does the intended application have reusable assets?
Affine value function	How much state dependence is lost within one solve?
Algorithm and scaling	Which claims are proved, certified, or only observed on test instances?
Published policy comparison	How much gain comes from time prices, lookahead, or re-solving?

Final evidence rule: compare bounds with exact values where possible; compare policies using exact evaluation or paired simulation. Label published results separately from project measurements.

Source: Reviewer-style questions derived from Adelman (2007), §§2–5; empirical answers remain pending.

Adelman and individualized pricing: a research lineage

Distinct mechanisms

Adelman (2007) estimates an *internal capacity opportunity cost*. Personalized pricing also models how an individual responds to an offered price.

Research thread	Candidate references supplied for verification
Customer choice and related methods	Zhang and Adelman (2009); Barz et al. (2023); Hausenblas et al. (2025)
Individualized pricing	Jahandideh et al. (2020); Lei et al. (2023/2025)
Parallel personalization line	Acquisti and Varian (2005); Aydin and Ziya (2009); Ban and Keskin (2021)
Policy context	FTC inquiry (January 2025): consumer data in individualized pricing

Evidence boundary: these are reading leads, not yet source-checked for this project. A citation trail does not establish that Adelman created surveillance-pricing systems.

★ **Decision:** Is a conceptual, source-checked connection sufficient for the final brief?

Source: Adelman (2007), pp. 647–648; subsequent works and FTC item supplied by the researcher, pending source verification.

IPR 3 gates and decisions requested

Next evidence gate	Source-linked proof notes; exact DP and LP agreement; fluid/static tests; full small affine LP; pricing versus exhaustive search; paired policy comparisons.
★ Proof (8)	Is the theorem and LP treatment sufficient for the final brief?
★ Algorithm (10)	How much pricing and column generation must IPR 3 implement?
★ Evidence (11)	Which published result and comparison protocol should IPR 3 target?
★ Application (13)	Is a conceptual, source-checked connection sufficient?

Source: ISE 631 course prompt, p. 2; project phase plan.

A1. Full transition and Bellman model

Let $\mathcal{U}(r) = \{u \in \{0,1\}^n : a_{ij}u_j \le r_i \ \forall i, j\}$, $p_{t0} = 1 - \sum_j p_{tj}$, and $q_t(r,u) = \sum_j p_{tj} f_j u_j$. Then

$$P_t(r' \mid r, u) = p_{t0}\mathbf{1}\{r' = r\} + \sum_j p_{tj}\mathbf{1}\{r' = r - A_j u_j\}.$$

Overlapping next states add their probabilities. With $v_{T+1} \equiv 0$,

$$v_t(r) = \max_{u \in \mathcal{U}(r)} \left\{ q_t(r,u) + \sum_{r'} P_t(r' \mid r, u) v_{t+1}(r') \right\}.$$

Initial state is c in period 1. No discount factor applies to this finite-horizon model.

Variables: t, T period and horizon; r, r' current and next resource vectors; c initial vector; i, j, n resource, class, class count; a_{ij}, A_j class resource use; u_j, $\mathcal{U}(r)$ accept indicator and feasible actions; p_{tj}, p_{t0} arrival/no-arrival probabilities; f_j fare; q_t expected reward; P_t transition; v_t optimal value; $\mathbf{1}$ indicator.

Source: Adelman (2007), pp. 648–649, eq. (1); transition notation derived.

A2. Exact finite-horizon Bellman LP and occupancy dual

Primal (D0): $\min v_1(c)$, with $v_t(r) \in \mathbb{R}$, $v_{T+1} = 0$, and

$$v_t(r) - \sum_{r'} P_t(r' \mid r, u) v_{t+1}(r') \geq q_t(r, u) \quad \forall t, r \in \mathcal{R}_t, u \in \mathcal{U}(r).$$

Derived dual: $x_t(r, u) \geq 0$ and

$$\max \sum_{t,r,u} q_t(r, u) x_t(r, u)$$

subject to

$$\sum_u x_1(c, u) = 1, \qquad \sum_u x_t(r, u) = \sum_{s,u} P_{t-1}(r \mid s, u) x_{t-1}(s, u) \quad \forall t = 2, \ldots, T, \ r \in \mathcal{R}_t.$$

Strong duality gives the exact initial value. Positive $x_t(r, u)$ makes its Bellman inequality tight.

Variables: t, T period/horizon; r, s states in $\mathcal{R}_t$; $u \in \mathcal{U}(r)$ feasible action; c initial state; q_t expected reward; P_t transition probability; v_t free value variable; x_t nonnegative state-action occupancy; $v_{T+1} = 0$ terminal value.

Source: Adelman (2007), p. 649, eq. (3); dual independently derived, not labeled P0 in the paper.

A3. Instructor's discounted MDP LP

For finite states S, actions $A(s)$, discount $0 \le \gamma < 1$, and initial weight $\nu \ge 0$:

$$\begin{aligned} \min_{V \in \mathbb{R}^S} \quad & \sum_s \nu(s) V(s) \\ \text{s.t.} \quad & V(s) - \gamma \sum_{s'} P(s' \mid s, a) V(s') \ge r(s,a) \quad \forall s, a \in A(s). \end{aligned}$$

The occupancy dual maximizes $\sum_{s,a} r(s,a) x(s,a)$ over $x \ge 0$ with

$$\sum_a x(s,a) - \gamma \sum_{s',a'} P(s \mid s', a') x(s', a') = \nu(s) \quad \forall s.$$

x counts ν-weighted discounted visits. Complementary slackness identifies optimal actions on occupied states. With $\nu(s) > 0$ for all s, the minimizing value vector is V^* everywhere.

Variables: S state set; $A(s)$ feasible actions; s, s' states; a action; γ discount; ν initial state weights; V, V^* candidate/optimal values; P transition probability; $r(s,a)$ reward; $x(s,a)$ discounted occupancy.

Source: ISE 631 instructor prompt, p. 1; dual and full-support qualification derived.

A4. Fluid primal, dual, and policy distinction

Let $d_j = \sum_t p_{tj}$. The fluid primal is

$$\max_{Y \geq 0} \sum_j f_j Y_j \quad \text{s.t. } AY \leq c, \quad Y_j \leq d_j \ \forall j.$$

Its dual is

$$\min_{\lambda, \mu \geq 0} \ c^\top \lambda + d^\top \mu \quad \text{s.t. } A_j^\top \lambda + \mu_j \geq f_j \ \forall j.$$

Capacity shadow price λ_i is nonnegative. Complementary slackness characterizes primal-dual optima; an online rule accepting when $f_j \geq A_j^\top \lambda$ is a policy constructed from these prices, with separate performance to be evaluated.

Variables: t period; j class; $d_j = \sum_t p_{tj}$ expected demand; Y_j expected acceptances; f_j fare; $A = (a_{ij})$ resource use; c initial capacity; λ_i resource dual; μ_j demand-cap dual.

Source: Adelman (2007), p. 650, eqs. (4)–(6) and displayed dual.

A5. Full affine Bellman LP (D1)

With $\mathcal{R}_1 = \{c\}$, $\mathcal{R}_t = \mathcal{R}$ for $t = 2, \ldots, T$, and $\theta_{T+1} = 0$, $b_{T+1} = 0$, the raw affine coefficients $\theta_t, b_{t,i} \in \mathbb{R}$ are free. The program is

$$\min_{\theta, b} \quad \theta_1 + b_1^\top c$$

subject to

$$\theta_t - \theta_{t+1} + (b_t - b_{t+1})^\top r \geq \sum_j p_{tj}(f_j - b_{t+1}^\top A_j)u_j$$

$$\forall t = 1, \ldots, T, \quad r \in \mathcal{R}_t, \quad u \in \mathcal{U}(r).$$

Theorem 2 permits a nonnegative monotone optimal selection only under its stated positive-capacity and positive-arrival assumptions.

Variables: $t = 1, \ldots, T$ period; $r \in \mathcal{R}_t$ remaining state; c initial capacity; $u \in \mathcal{U}(r)$ feasible acceptance vector; θ_t free intercept; b_t free resource slopes; p_{tj} arrival probability; f_j fare; A_j class-j resource use.

Source: Adelman (2007), p. 652, eqs. (14), (16)–(17); Theorem 2, pp. 654–655.

A6. Affine dual and the bound proof

For (P1), $X_t(r,u) \geq 0$ maximizes $\sum_{t,r,u} q_t(r,u) X_t(r,u)$ subject to

$$\sum_{r,u} X_t(r,u) = 1 \quad \forall t,$$

$$\sum_{r,u} r_i X_1(r,u) = c_i, \quad \sum_{r,u} r_i X_t(r,u) = \sum_{s,u} \left(s_i - \sum_j p_{t-1,j} a_{ij} u_j \right) X_{t-1}(s,u) \quad \forall i, t > 1.$$

These are moments, not statewise flows. Set $Y_j = \sum_{t,r,u} p_{tj} u_j X_t(r,u)$. Mass gives $Y_j \leq d_j$. Telescoping resource moments gives

$$c_i = \sum_{t<T,r,u,j} p_{tj} a_{ij} u_j X_t(r,u) + \sum_{r,u} r_i X_T(r,u) \geq \sum_j a_{ij} Y_j,$$

since $r_i \geq \sum_j p_{Tj} a_{ij} u_j$. Therefore every P1 objective is fluid feasible. D1 feasibility separately gives $z^{P1} \geq v_1(c)$.

Variables: t, T period/horizon; i, j resource/class; r, s remaining states; u_j accept indicator; $X_t(r,u)$ P1 weight; q_t expected reward; p_{tj} arrival probability; a_{ij} resource use; c_i initial capacity; Y_j, d_j expected acceptances/arrivals; z^{P1}, $v_1(c)$ affine bound/exact value.

Source: Adelman (2007), pp. 652–653, eqs. (18)–(24), Theorem 1.

A7. Monotone prices: scope of Theorem 2

Under $c_i > 0$ and $p_{tj} > 0$ for every i, t, j, an optimal D1 solution exists with

$$b_{t,i} \geq b_{t+1,i} \geq 0, \qquad \theta_t \geq \theta_{t+1} \geq 0.$$

The proof uses the first period in which a resource appears in positive P1 support, complementary slackness, and changes to the dual prices that preserve feasibility and optimality.

Why "exists" matters: Example 1 also admits $\theta_2 = 0$, $b_2 = 0.5$, $\theta_1 = 0.75$, $b_1 = 0$. Its D1 objective is 0.75, but $b_1 < b_2$. Zero-demand or zero-capacity tests lie outside the theorem's stated assumptions.

Variables: t period; i resource; j class; c_i initial capacity; p_{tj} arrival probability; $b_{t,i}$ affine resource price; θ_t affine intercept; D1/P1 are the affine primal-dual pair.

Source: Adelman (2007), pp. 653–655, eqs. (25)–(29), Theorem 2 and Corollary 1; counterexample derived.

A8. Pricing, reduced costs, and stopping

For fixed $t > 1$, define $w_j = p_{tj}(f_j - b_{t+1}^{\top} A_j)$ and $d_i = b_{t,i} - b_{t+1,i}$. Adelman's pricing problem is

$$\max_{r,u} \sum_j w_j u_j - \sum_i d_i r_i - \theta_t + \theta_{t+1} \quad \text{s.t. } a_{ij} u_j \le r_i, \quad u_j \in \{0,1\}, \quad 0 \le r_i \le c_i.$$

The paper uses an integer program and generates a batch of improving columns. With global period maxima ρ_t^*, Proposition 3 yields $z^{P1} \le Z_\Lambda + \sum_t \rho_t^*$; if $Z_\Lambda > 0$, $\sum_t \rho_t^* / Z_\Lambda \le \varepsilon$ is a sufficient relative stopping rule.

Complexity caveat: A derived minimum-cut reduction for this exact pricing structure is being proof-checked. The paper itself does not claim NP-hardness or polynomial complexity.

Variables: t, T period/horizon; i, j resource/class; w_j net expected class profit; d_i price change; r_i, c_i remaining/initial capacity; a_{ij} resource use; u_j accept indicator; θ_t, b_t intercept/slopes; ρ_t^* max reduced profit; Z_Λ restricted-primary value; z^{P1} full value; ε tolerance.

Source: Adelman (2007), pp. 650–651, Proposition 3, and pp. 655–657, eqs. (33)–(36), Figure 2.

A9. Proposed experiment matrix

Block	Variation	Evidence to record
Verification	Example 1; 2-resource case; edge cases	DP, exact LP and dual, fluid, full affine LP; values and residuals
Bound strength	Size, load, timing, fare	Exact V^* when feasible; fluid and certified affine bounds
Policy value	Fixed and re-solved policies	Tiny-case exact value; policy and paired-difference 95% intervals
Computation	Enumeration vs. columns	Runtime, memory, iterations, generated columns, pricing gaps
Truncation	Columns, states, horizon	Time saved, omitted constraints, surviving certificates

Factor levels, pilot size, final replication precision, seeds, and published replication target require approval before Phase 4 runs.

Source: Project experiment design proposal; no project experimental results.

A10. Instructor requirements I01–I07

ID	Final part	Register	Evidence needed next
I01	2–4	Planned	Discounted Bellman LP, occupancy dual, complementary slackness audit
I02	2	Planned	Exact network MDP and state-growth analysis
I03	2–3	Planned	Fluid primal/dual and static-policy validation
I04	3–5	Planned	Source-matched affine ALP and algorithm audit
I05	3	Planned	Paper method and solution approach, source-linked
I06	3–5	Planned	Correct LP course-concept mapping
I07	4	Planned	Principal theorem and independently reviewed proof

Draft formulas in this brief are presentation material. The register remains planned until evidence paths and a separate reviewer are recorded.

Source: `requirements.csv`, I01–I07; instructor prompt, pp. 1–2.

A11. Instructor requirements I08–I14

ID	Final part	Register	Evidence needed next
I08	6	Planned	Reproducible fluid, static, and dynamic policy tests
I09	3–4	Planned	Approximation benefit with verified example
I10	4	Planned	Full bound-dominance derivation and audit
I11	5	Planned	Pricing formulation and qualified complexity account
I12	6	Planned	Policy, exact MDP, and fluid gap results
I13	7	Planned	Evidence-based critique and alternatives
I14	1, 5, 8	Planned	Approved independent RQs linked to final analysis

I14 has approved questions on file, but its requirement register is not reviewed. I08 and I12 have no project numerical results yet.

Source: `requirements.csv`, I08–I14; instructor prompt, p. 2.